Combinatorial geometry

L4-X5 Level 4 Olympiad · Combinatorics · Beyond the textbook

Kevin Hu
November 2026

Convex hulls and extremal choices, pigeonhole on regions, lattice points and parity, counting crossings, and covering arguments for olympiad problems about finite point sets.
By the end you can choose an extremal object (hull vertex, closest pair, largest triangle) and finish a proof from that object; design pigeonhole partitions of a region, use parity and Pick's theorem on lattice points, and count crossings by double counting; prove that a shape can be placed avoiding obstacles by enlarging them and comparing areas.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1). **Contests:** BMO1, BMO2, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Combinatorial geometry asks what the arrangement of a finite set of points forces. The subject comes from the Hungarian school. Esther Klein's five-point question led to the 1935 paper of Erdős and Szekeres. Textbooks skip this history because there is nothing to calculate. We assume L4-CO1.

1.1 Extremal choices and the convex hull

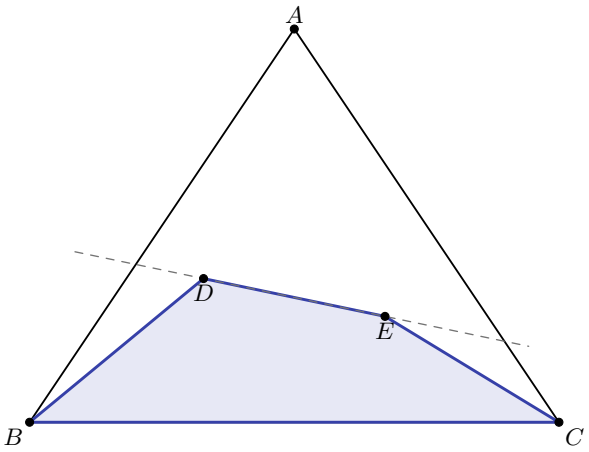
Theorem 1 (convex hull).
Let S be a finite set of points, not all on a line. The smallest convex set containing S is a convex polygon, the *convex hull*, with vertices in S. A point P of S is a hull vertex if and only if some line through P has all other points of S in one open half-plane. If AB is a hull side, S lies in one closed half-plane of line AB.

We use these without proof. Other extremal objects are the closest pair, the largest or smallest triangle.

Theorem 2 (Klein).
Any five points in the plane, no three collinear, include the vertices of a convex quadrilateral.

Proof. If the hull has four or five vertices, take four of them. Otherwise the hull is a triangle ABC containing D and E. Line DE meets two sides, so the line separates one vertex from the other two, say A from B and C. If one of B, C, D, E lay inside the triangle of the other three, the line through that point and the opposite vertex would cross the opposite side. For D inside BCE this separates B from C by line DE, and for B inside CDE the line separates D from E by line BC (the cases of E and C are the same). So B, C, D, E are in convex position. ■

Kevinvariant
kevinvariant08.github.io



Theorem 3 (largest triangle).
If every triangle with vertices in a finite set S has area at most 1, then S lies in a triangle of area 4.

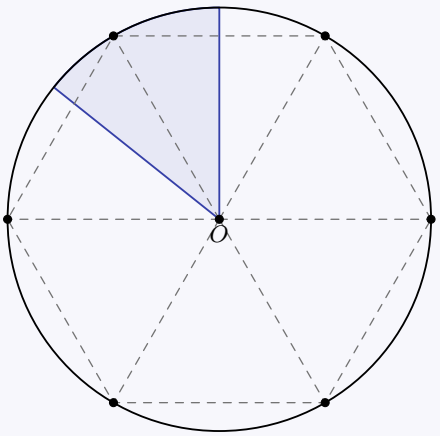
Proof. (Extremal choice.) Let ABC have the largest area. The lines through A, B, C parallel to the opposite sides form a triangle T with $[T] = 4[ABC] \leq 4$. A point of S beyond the line through A parallel to BC would form with B, C a triangle larger than ABC. So $S \subseteq T$.

1.2 Pigeonhole on a region

Principle.
If a region is cut into k pieces of diameter at most d, then among any $k+1$ points of the region two are at distance at most d.

Find the extremal configuration first and draw one piece around each of its points.

Example 1.
Eight points lie in a closed disc of radius 1. Prove that two of them are at distance less than 1.



Solution.

Kevinvariant
kevinvariant08.github.io

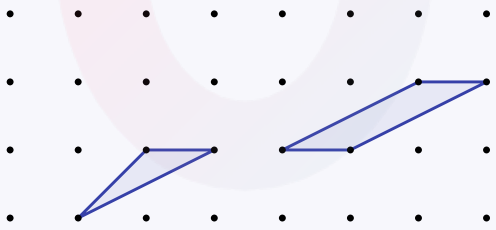
Seven sectors of angle $\frac{360^\circ}{7}$ have diameter 1, so pieces give only "at most 1" (the centre and a point of the arc are at distance 1).
Use angles instead. Let O be the centre. If $P, Q \neq O$ and $\angle POQ < 60^\circ$, then another angle of triangle OPQ exceeds 60°, so $PQ < \max(OP, OQ) \leq 1$ (if $\angle POQ = 0^\circ$, $PQ = |OP - OQ|$). At most one point is O, and the rays to the other seven include two at angle at most $\frac{360^\circ}{7} < 60^\circ$. ■
Remark. The centre and a regular hexagon of side 1 show that seven points do not suffice.

1.3 Lattice points and parity

Theorem 4 (parity classes).
Among any five lattice points in the plane, two have a lattice midpoint, and in space among any nine.
Theorem 5 (Pick, 1899). A simple polygon with lattice vertices, I lattice points inside and B on its boundary has area $I + \frac{1}{2}B - 1$. So a lattice triangle with no lattice points except its vertices has area $\frac{1}{2}$.

The midpoint of (a, b) and (c, d) is a lattice point if and only if $a \equiv c$ and $b \equiv d \pmod 2$, and there are four classes. For Pick's theorem see Aigner and Ziegler, *Proofs from THE BOOK.* , "Three applications of Euler's formula". The shoelace formula alone puts every lattice area in $\frac{1}{2}\mathbb{Z}$, so no equilateral triangle has lattice vertices, as its area $\frac{\sqrt{3}}{4}s^2$ with $s^2 \in \mathbb{Z}$ is irrational.

Example 2.
A convex polygon has its vertices at lattice points and contains no other lattice points, inside or on its boundary. Prove that the polygon is a triangle of area $\frac{1}{2}$ or a quadrilateral of area 1.



Solution.
With five or more vertices, Theorem 4 gives two with a lattice midpoint M, which lies in the polygon by convexity. M is not a vertex, since a line through a vertex V meets the polygon only at V, and two points of the polygon with midpoint V would lie on that line, hence equal V. So there are at most four vertices, and Pick with $I = 0$, $B = 3$ or 4 gives area $\frac{1}{2}$ or 1. ■
Remark. Lattice points and the number five suggest checking parity classes first.

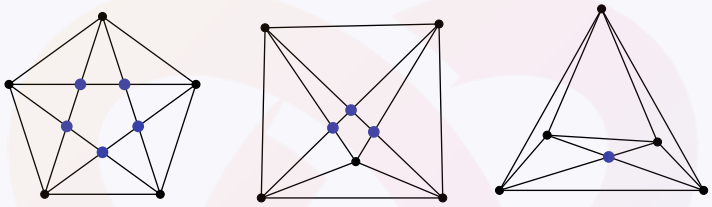
Kevinvariant
kevinvariant08.github.io

1.4 General position and counting crossings

Theorem 6 (crossings).
Join every pair of points of a set S with no three collinear (in *general position*). Two segments cross if and only if their endpoints are four distinct points in convex position and the segments are its diagonals. So the number of crossing pairs is the number of convex quadrilaterals in S, which is $\binom{n}{4}$ for n points in convex position.

Proof. Segments with a common endpoint meet only there. In a convex quadrilateral only the diagonals cross. If D is inside triangle ABC, the segments from D meet the triangle's boundary only at A, B, C, so nothing crosses.

Example 3.
Five points in general position are joined in pairs by all ten segments. Find all possible numbers of regions into which the segments cut the plane.



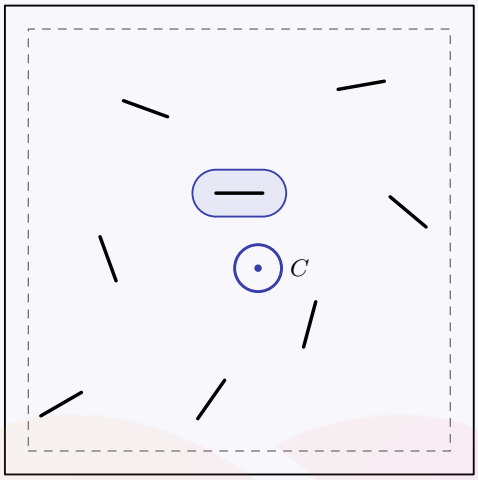
Solution.
Let X be the number of crossing pairs. The four segments from one point form a connected drawing with one region. A further segment crossing c earlier ones is cut into $c+1$ pieces, each joining two points of the drawing and splitting a region. So there are $1+6+X=7+X$ regions.
By Theorem 6, X counts convex quadrilaterals. A pentagonal hull gives $X=5$. A hull $ABCD$ with E inside gives $X=3$, because $ABCD$ is convex, and E lies in one of ABC, ACD and one of ABD, BCD, so two quadruples containing E are not convex. A hull ABC with D, E inside gives $X=1$, because a convex quadruple must contain D and E, and for the two vertices that line DE separates, DE is neither a side nor a diagonal. The answer is 12, 10 or 8.
Remark. For n points the same count gives $\binom{n}{2}-n+2+X$ regions.

1.5 Covering arguments

Enlargement.
Fix a reference point of a shape. The positions of this point at which the shape meets an obstacle form an enlarged obstacle, and if these have total area less than the region of allowed positions, some position is free.
Area pigeonhole. If sets inside a region of area A have total area greater than kA, some point lies in at least $k+1$ of them, since otherwise adding the areas counts each point at most k times.

Kevinvariant
kevinvariant08.github.io

Example 4.
Forty-five matches, each a segment of length 1, lie on a square table of side 10. Prove that a coin of diameter 1 can be placed flat on the table, without overhanging the edge, so that the coin touches no match.



Solution.
The coin's centre C must lie in the concentric square of side 9, of area 81. The coin touches a match if and only if C is within $\frac{1}{2}$ of that match, that is, in a stadium of area $1+\frac{\pi}{4}$. Since $\pi < 3.2$,

$$45\left(1+\frac{\pi}{4}\right) < 45+36 = 81,$$

so some point of the square lies in no stadium. ■

Common traps.

- Pieces give "at most d", not "less than d".
- Claiming a bound is best possible without the configuration.
- Assuming general position when general position is not given.
- Letting the reference point range over the whole table instead of the smaller region.

1.6 How to spot this in a contest

1. Points in a region and a distance to bound suggest pigeonhole on pieces, or on angles about a centre.
2. "Some point (triangle, line) of a finite set has a property" suggests a hull vertex, the closest pair or the largest triangle.
3. Lattice points suggest parity classes, areas in $\frac{1}{2}\mathbb{Z}$, Pick, rational slopes.
4. Counting crossings or regions suggests a fact about four or five points, then double counting.
5. "Can be placed so as to avoid" suggests enlarging the obstacles and compare areas.

Kevinvariant
kevinvariant08.github.io

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Nine points lie inside or on a cube of edge 2. Prove that two of them are at distance at most $\sqrt{3}$, and show that $\sqrt{3}$ cannot be replaced by any smaller number.

Problem 2 ⋆⋆ A polygon has its vertices at lattice points. Prove that if one of its angles is a whole number multiple of 15°, then that angle is a multiple of 45°.

Problem 3 ⋆⋆ No three diagonals of a convex polygon pass through one point inside the polygon. The points where diagonals cross cut the diagonals into 455 segments in total. How many sides does the polygon have?

Problem 4 ⋆⋆⋆ A shape cut from card has area $10\frac{1}{2}$. A sheet of paper is ruled into unit squares. Prove that the shape can be laid on the sheet, without rotating the shape, so that the shape covers at least 11 vertices of the grid, and also so that the shape covers at most 10.

Problem 5 ⋆⋆⋆ Finitely many points lie in the plane, and every two of them are at distance at least 1. Prove that one of them is at distance exactly 1 from at most three of the others. Show that three cannot be replaced by two.

Problem 6 ⋆⋆⋆⋆ A convex quadrilateral $ABCD$ has its vertices at lattice points. Its sides contain no lattice points other than A, B, C, D, and exactly one lattice point P lies inside the quadrilateral. Prove that P is the midpoint of AC or of BD.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1994 Problem 7** ⋆⋆
   List the lattice points on $x^2 + y^2 = 50$, then think about which lines through them or tangent to the circle are allowed.
2. **Japan MO 1997 (AoPS wiki, Pigeonhole Principle, Olympiad Problems)** ⋆⋆
   Cut the disc into nine regions, each of diameter less than 2, and count.
3. **Baltic Way 1992 Problem 4** ⋆⋆⋆
   Write each squared side as a sum of two squares of integers and track parities around the hexagon.
4. **USAMTS Year 18 Round 1 Problem 4** ⋆⋆⋆
   Pigeonhole on a suitable finite grid of points, first on one line, then on several parallel lines.
5. **Baltic Way 2003 Problem 10** ⋆⋆⋆
   Reduce coordinates modulo 4 and ask how many points of a given residue class you can keep.
6. **BMO2 2020/21 Problem 2** ⋆⋆⋆
   *Think about which side lengths n can be written as a*x+b*y in two ways, and what that does to the tiling.*

Kevinvariant
kevinvariant08.github.io

7. **IMO Longlist 1959-66 Czechoslovakia 1** ⋆⋆⋆
   Start from an edge of the convex hull and choose the third point extremally.

8. **BMO1 2017/18 Problem 5** ⋆⋆⋆
   Look for many disjoint small regions of the 200 x 200 board, each of which must contain a coloured square to destroy every comb inside that region.

9. **IMO 1989 Problem 3** ⋆⋆⋆⋆
   Count pairs (point, pair of points equidistant from that point) in two different ways.

10. **BMO2 2014/15 Problem 4** ⋆⋆⋆⋆
    Try small cases of an n-loop for the first few n, then look for a way to extend a loop by two points.

11. **IMO 1969 Problem 5** ⋆⋆⋆⋆
    First show that among any five points, no three collinear, some four form a convex quadrilateral, then count by double counting.

Kevinvariant
kevinvariant08.github.io

4 Hints

1. Cut the cube into eight pieces. For the second part, use the most symmetric nine points you can find.
2. Write the tangent of the angle between the lattice vectors (a, b) and (c, d) in terms of a, b, c, d.
3. Each crossing point is determined by four vertices. How many extra segments does one crossing point create?
4. Cut the shape along the grid lines and slide every piece into one unit square. What does a point covered by several pieces correspond to?
5. Look at a vertex of the convex hull. What is the smallest angle two of its neighbours can make at that vertex?
6. Find the areas of the four triangles PAB, PBC, PCD, PDA. What does $[PAB] = [PBC]$ say about line BP?

5 Solutions

Problem 1

Nine points lie inside or on a cube of edge 2. *Prove that two of them are at distance at most* $\sqrt{3}$, *and show that* $\sqrt{3}$ *cannot be replaced by any smaller number.*

The three planes through the centre parallel to the faces cut the cube into eight cubes of edge 1, and a point on a cutting plane is assigned to any cube containing the point. Two of the nine points lie in the same small cube, and a cube of edge 1 has diameter equal to its space diagonal $\sqrt{3}$.

For the eight vertices and the centre, two vertices are at distance at least 2 and the centre is at distance $\sqrt{3}$ from each vertex, so the least distance is $\sqrt{3}$. This configuration is also where the pieces came from, namely one small cube around each vertex, with the centre shared by all eight.

Answer: $\sqrt{3}$ is attained by the eight vertices and the centre

Problem 2

A polygon has its vertices at lattice points. Prove that if one of its angles is a whole number multiple of 15°, *then that angle is a multiple of* 45°.

The two sides at a vertex point along lattice vectors $u = (a, b)$ and $v = (c, d)$. The angle θ between them satisfies $|u||v| \cos\theta = ac + bd$ and $|u||v| \sin\theta = |ad - bc|$, so

$$\tan\theta = \frac{|ad - bc|}{ac + bd}$$

is rational unless $ac + bd = 0$, when $\theta = 90^\circ$. The angle φ of the polygon is θ or $360^\circ - \theta$, so $\tan\varphi = \pm\tan\theta$ is rational or $\varphi \in \{90^\circ, 270^\circ\}$.

A multiple of 15° that is not a multiple of 45° differs from 15°, 30°, 60° or 75° by a multiple of 180°, possibly after changing sign, and these have tangents $2 - \sqrt{3}$, $\frac{1}{\sqrt{3}}$, $\sqrt{3}$, $2 + \sqrt{3}$, all irrational. So such an angle has an irrational tangent and cannot occur. ■

Kevinvariant
kevinvariant08.github.io

In particular no lattice triangle is equilateral, a second proof of the fact in the lesson. The angles $45°$, $90°$ and $135°$ all occur.

Problem 3

No three diagonals of a convex polygon pass through one point inside the polygon. The points where diagonals cross cut the diagonals into 455 *segments in total. How many sides does the polygon have?*

Let the polygon have n sides. By Theorem 6 every four vertices give exactly one crossing point, the crossing of their diagonals, and as no three diagonals are concurrent different quadruples give different points. So there are $\binom{n}{4}$ crossing points. Each lies on exactly two diagonals and adds one segment to each.

The polygon has $\frac{n(n-3)}{2}$ diagonals, so the number of segments is

$$\frac{n(n-3)}{2} + 2\binom{n}{4}.$$

This increases with n. For $n = 9$ the number is $27 + 252 = 279$ and for $n = 10$ the number is $35 + 420 = 455$. The polygon has 10 sides.

Answer: 10

Problem 4

A shape cut from card has area $10\frac{1}{2}$*. A sheet of paper is ruled into unit squares. Prove that the shape can be laid on the sheet, without rotating the shape, so that the shape covers at least* 11 *vertices of the grid, and also so that the shape covers at most* 10*.*

Put the grid vertices at the lattice points and let $Q = [0,1)^2$. Cut the shape S into the pieces $S \cap (m + Q)$ for lattice points m, and translate each piece by $-m$ into Q. A point t of Q lies in the translated piece from m if and only if $t + m \in S$, that is, if and only if the lattice point m is covered by the shape moved by $-t$. So the number $N(t)$ of translated pieces containing t is the number of grid vertices covered by $S - t$.

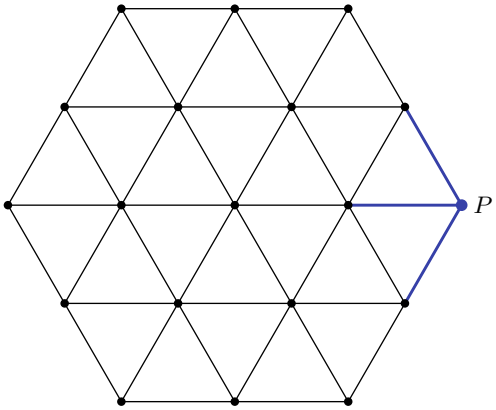
The pieces have total area $10\frac{1}{2} > 10 \cdot [Q]$, so by area pigeonhole $N(t) \geq 11$ for some t. If instead $N(t) \geq 11$ for every t, the pieces would have total area at least 11. So $N(t) \leq 10$ for some t, and the two translates $S - t$ are the required placings. ■

The average of $N(t)$ over Q is the area of S. The case of area greater than 1 is Blichfeldt's lemma (1914), which says that some two points of S differ by a lattice vector.

Problem 5

Finitely many points lie in the plane, and every two of them are at distance at least 1*. Prove that one of them is at distance exactly* 1 *from at most three of the others. Show that three cannot be replaced by two.*

Kevinvariant
kevinvariant08.github.io



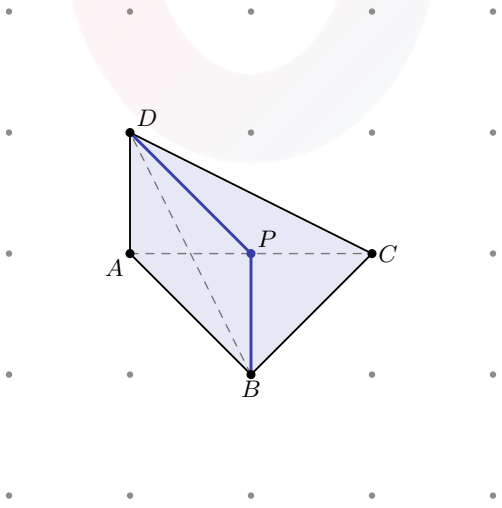
If all the points are on a line, the two end points each have at most one neighbour at distance 1. Otherwise let P be a vertex of the convex hull. Its angle in the hull is less than $180°$, and every other point lies inside that angle or on its arms. Let $Q_1, \ldots, Q_k$ be the points at distance 1 from P, in order of angle about P.

For $i \neq j$, triangle PQ_iQ_j has $PQ_i = PQ_j = 1 \leq Q_iQ_j$, so its largest angle is at P and $\angle Q_iPQ_j \geq 60°$. Hence the angle Q_1PQ_k is at least $60°(k-1)$, and this angle is less than $180°$, so $k \leq 3$. ■

For the second part take the 19 points of the triangular lattice with spacing 1 that lie in a regular hexagon of side 2. All distances are at least 1, the six corners have three neighbours at distance 1 each and every other point has at least four.

Problem 6

A convex quadrilateral $ABCD$ has its vertices at lattice points. Its sides contain no lattice points other than A, B, C, D, and exactly one lattice point P lies inside the quadrilateral. Prove that P is the midpoint of AC or of BD.



The triangle PAB contains no lattice points other than its vertices, since its other points lie on side AB or inside the quadrilateral. By Pick's theorem $[PAB] = \frac{1}{2}$, and in the same way $[PBC] = [PCD] = [PDA] = \frac{1}{2}$.

Line BP meets the boundary again at a point of side CD or DA, so this line separates A from C. Since $[PAB] = [PBC]$, the points A and C are at equal distances from line BP, so the line passes

Kevinvariant
kevinvariant08.github.io

through the midpoint M of AC. From $[PCD] = [PDA]$, line DP also passes through M. In the same way lines AP and CP pass through the midpoint N of BD.

Suppose $P \neq M$ and $P \neq N$. Then lines BP and DP are both the line PM, so P and M lie on BD, and likewise P and N lie on AC. So P lies on both diagonals and is their crossing point, and M, lying on both diagonals, is that point too. This gives $P = M$, a contradiction. ■

Both cases occur. For $A = (0,1)$, $B = (1,0)$, $C = (2,1)$, $D = (0,2)$ the point $P = (1,1)$ is the midpoint of AC but not of BD, while for a parallelogram P is the midpoint of both. The same argument shows that if a lattice triangle has exactly one lattice point inside and none on its sides other than its vertices, that point is its centroid.